TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

Proposer	Name	Mohammad Mohammadi Amiri	Email	mamiri@rpi.edu
Institute Name	Rensselaer Polytechnic Institute			
Address	110 8th Street, Troy, NY 12180, USA			

2. Technology Overview

Title of Proposed Technology	Scalable Multi-Agent Guardrail Technology for Safe AI Systems			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication			
	Medium Category	Small Category	Medium Category	Small Category
	Software	Embedded Software		
Concept of Proposed Technology	The proposed technology is a multi-agent AI safeguarding infrastructure designed to ensure safe and reliable deployment of large language models in real-world applications. Its core function is to monitor, evaluate, and control model outputs using specialized guardrail agents coordinated by a central safety orchestrator. The technology enables real-time detection and mitigation of harmful content such as toxicity, bias, misinformation, and privacy violations. It can be applied in AI-powered products, including virtual assistants, enterprise analytics systems, customer support platforms, and decision-support tools requiring reliable and trustworthy AI behavior.			
Rationale for Joint R&D Project	• Small and medium-sized companies face growing challenges in deploying large AI models safely and reliably due to limited internal expertise and high system complexity. • Existing AI safety and moderation solutions are often difficult to integrate into production environments and lack flexibility for diverse real-world applications. • There is a strong need for practical, scalable safeguarding infrastructure that enables trustworthy AI deployment while meeting performance and cost constraints. • The research institute contributes specialized expertise in multi-agent AI systems, safety algorithms, and system-level optimization that is typically unavailable within SMEs. • The Korean SME partner may provide real-world application scenarios, deployment environments, and commercialization pathways essential for practical validation and adoption. • Joint R&D enables rapid prototyping, iterative refinement based on operational feedback, and faster transition from research outcomes to commercial products. • Collaboration reduces technical risk, shortens development timelines, and ensures alignment between advanced research and market-driven requirements.			
Key Words	1. Multi-Agent Safeguarding 2. Safe and Reliable AI			

	3. AI Guardrails 4. Real-Time Content Moderation 5. Deployable AI Infrastructure

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 2	End of the Project	Level 4
R&D Objectives	• Develop a multi-agent AI safeguarding infrastructure that enables safe and reliable deployment of large language models in real-world applications. • Design and implement a central safety orchestrator that coordinates specialized guardrail agents and performs decision fusion, policy enforcement, and refinement control. • Create lightweight, specialized safeguarding agents for detecting and mitigating key risk types, including toxicity, bias, misinformation, and privacy violations. • Build a deployable software layer that integrates seamlessly with existing AI inference pipelines without requiring retraining of base models. • Develop real-time monitoring and intervention mechanisms that enable early detection, bounded refinement, and low-latency response during model generation. • Validate the technology through industry-relevant prototypes and pilot deployments to ensure robustness, scalability, and readiness for commercialization within three years.			

R&D Target Specifications	• Safeguarding Accuracy and Coverage
	  ○ Achieve ≥90% detection recall for major risk categories (toxicity, bias, misinformation, privacy) on representative benchmark and industry datasets.
	  ○ Maintain false positive rate ≤5–8% to minimize unnecessary blocking of benign content.
	• Inference Latency and Throughput Impact
	  ○ Limit additional end-to-end inference latency introduced by safeguarding to ≤10–20% compared to unprotected inference.
	  ○ Support real-time operation with per-token monitoring latency ≤50–100 ms in streaming generation settings.
	• Refinement and Intervention Effectiveness
	  ○ Achieve ≥80% successful correction rate for flagged harmful outputs through bounded refinement without requiring full regeneration.
	  ○ Limit average refinement iterations to ≤2 cycles per response to ensure responsiveness and stability.
	• Scalability and System Compatibility
	  ○ Support deployment on commercial GPUs and cloud-based inference platforms without hardware modification.
	  ○ Enable safeguarding of models up to 10–70B parameters with multi-agent orchestration overhead remaining bounded and predictable.
	• Deployment Readiness and Reliability
	  ○ Deliver a production-ready prototype capable of continuous operation with ≥99.5% system uptime in pilot deployments.
	  ○ Validate performance and reliability through at least one industry pilot system by the end of the project.			

R&D Plan	**Overall Approach and Methodology** The goal of this joint R&D project is to develop a practical, scalable, and deployable multi-agent AI safeguarding infrastructure that enables safe and reliable deployment of large AI models in real-world applications. The project adopts a three-year, application-driven R&D approach that progresses from core system architecture and prototype development to industry validation and commercialization readiness.			

Existing AI safeguarding approaches are predominantly based on monolithic classifiers, rule-based filters, or single-model moderation pipelines that operate independently of the generation process [1-3]. Although these methods have demonstrated effectiveness for detecting individual harm categories such as toxicity, bias, or misinformation in controlled benchmarks, they are typically static, difficult to extend across heterogeneous harm types, and poorly suited to real-time, interactive, or multi-step generation settings [4-7]. Recent work has highlighted that single-model guardrails suffer from limited coverage, calibration challenges, and vulnerability to adversarial prompting, while iterative self-refinement approaches risk instability and reward hacking [8-10]. Although recent research has explored multi-agent architectures for reasoning and control, there is currently no deployable safeguarding infrastructure that enables coordinated, modular, and real-time safety assessment through specialized agents in production environments [11]. This gap motivates the development of a multi-agent safeguarding system that provides scalable, extensible, and transparent guardrails for next-generation AI applications.

The overall methodology combines system-level architecture design, prototype-driven engineering, pilot-based validation, and progressive technology maturation. Rather than developing isolated detection models, the project focuses on building a modular safeguarding infrastructure in which multiple specialized guardrail agents collaborate under the coordination of a central Safety Orchestrator. This architecture enables flexible extension, efficient deployment, and transparent decision-making suitable for production environments.

Figure 1 illustrates the high-level concept of the proposed safeguarding workflow, showing how candidate outputs from a task model are evaluated and refined through coordinated agent-based assessment before being released to users.

The R&D plan is organized into three phases aligned with increasing technology maturity:
(1) core safeguarding architecture and prototype development,
(2) industry integration and validation, and
(3) commercialization readiness and scaling.

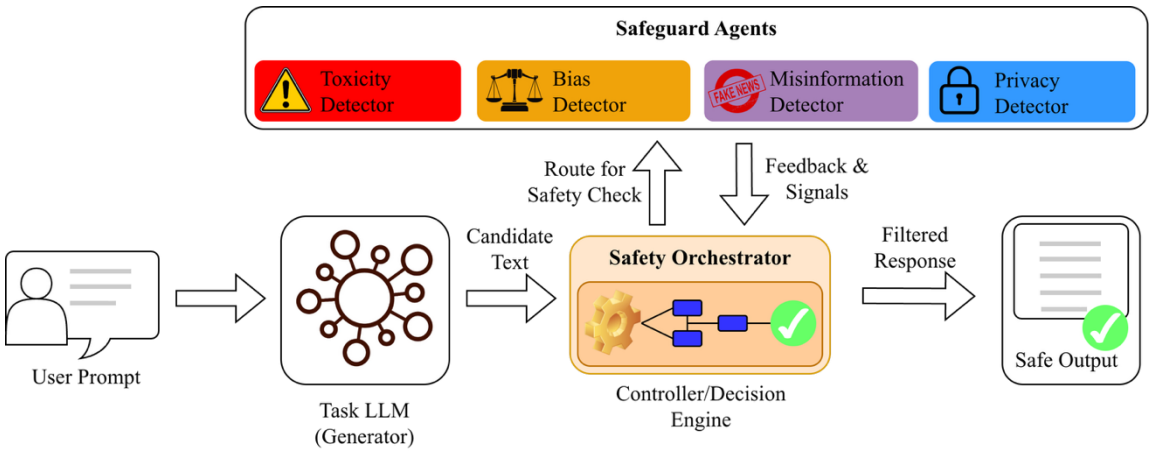

Figure 1. High-level workflow of the proposed multi-agent AI safeguarding framework, illustrating how candidate model outputs are evaluated and refined through coordinated guardrail agents before being released to users.

Year 1: Core Safeguarding Architecture and Prototype Development

Research Approach and Key Tasks

The first year focuses on establishing the foundational architecture of the multi-agent safeguarding framework and delivering an initial working prototype. The primary approach is architecture-driven system design combined with rapid prototyping, transforming the core research concepts into a

modular and deployable software system.

Key tasks in this phase include:

- Designing the overall safeguarding system architecture, including the roles of the Task LLM, specialized guardrail agents, and the Safety Orchestrator.
- Defining standard interfaces and communication protocols between agents to enable efficient coordination and extensibility.
- Implementing initial versions of specialized guardrail agents targeting representative harm categories such as toxicity, bias, misinformation, and privacy.
- Developing the Safety Orchestrator responsible for routing candidate outputs, aggregating agent signals, and producing final safeguarding decisions.

Initial evaluation methodologies will include controlled experiments on curated safety benchmarks and synthetic workloads to assess detection accuracy, latency, and system overhead.

Expected Milestones

By the end of Year 1, the project will deliver:

- A functional prototype of the multi-agent safeguarding architecture
- A baseline Safety Orchestrator coordinating at least three specialized guardrail agents
- Initial performance benchmarks characterizing detection quality and runtime cost

Figure 2 provides a zoom-in view of the Safety Orchestrator's decision fusion module, illustrating how multiple safety criteria are integrated by a policy layer to select allow, rewrite, or halt actions. The Safety Orchestrator applies decision fusion rules combining (i) threshold-based risk control, (ii) policy-based compliance logic, and (iii) multi-objective utility optimization, with optional extensions such as hierarchical priority rules, weighted voting, and context-aware fusion to balance safety, usefulness, and overblocking risk.

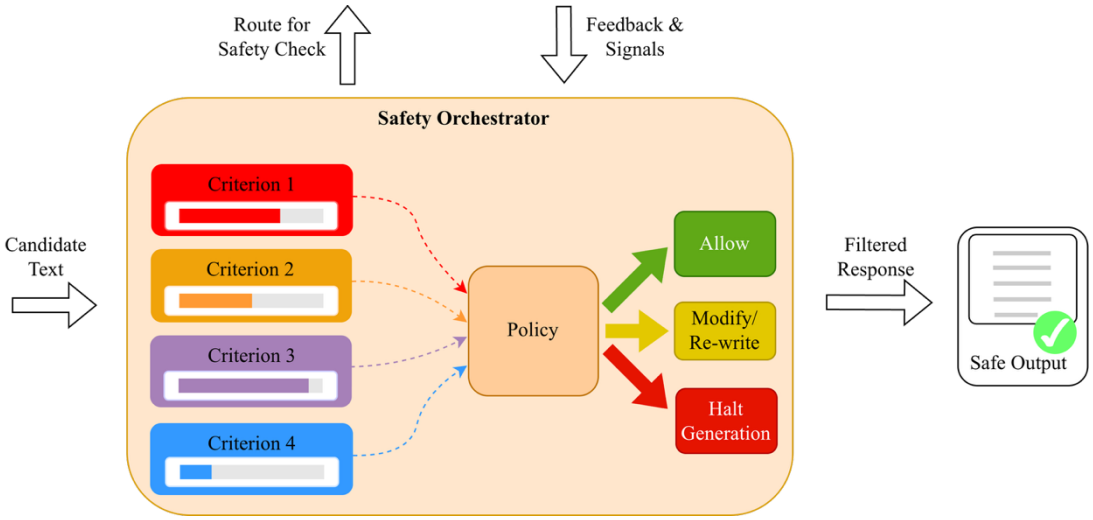

Figure 2. Decision fusion module within the Safety Orchestrator, combining multiple safety criteria to select allow, rewrite, or halt actions.

Year 2: Industry Integration and Validation

Research Approach and Key Tasks

The second year centers on pilot-driven system refinement and validation through close collaboration with the potential SME partner. The methodology emphasizes application-guided development, where real-world use cases shape system configuration, performance targets, and usability requirements.

Key tasks include:

- Integrating the safeguarding framework into industry-relevant AI applications such as interactive assistants, enterprise analytics tools, or content generation systems.
- Configuring deployment environments compatible with cloud, on-premise, or hybrid inference pipelines.
- Conducting pilot deployments under realistic operating conditions and collecting operational metrics on safety coverage, latency, false positives, and system robustness.
- Refining agent coordination strategies, decision fusion policies, and system configurations based on pilot feedback.

Evaluation will focus on end-to-end safeguarding effectiveness, including coverage across harm categories, runtime overhead, operational stability, and user experience impact.

Expected Milestones

By the end of Year 2, the project will deliver:

- Validated pilot deployments in at least one industry application
- Quantitative evaluation reports demonstrating improved safety coverage with acceptable latency and cost
- Refined orchestration and agent coordination mechanisms ready for production hardening

Year 3: Commercialization Readiness and Scaling

Research Approach and Key Tasks

The third year focuses on engineering maturation, scalability, and commercialization preparation. The methodology emphasizes system hardening, usability engineering, and deployment automation to ensure production readiness and ease of adoption by small and medium-sized companies.

Key tasks include:

- Improving system stability, fault tolerance, and monitoring capabilities for continuous operation.
- Developing configuration tools and management interfaces that allow non-expert users to deploy and customize safeguarding policies.
- Optimizing runtime performance and resource usage for large-scale or multi-tenant deployments.
- Working with industry partners to define commercialization pathways, product integration strategies, and go-to-market models.

Expected Milestones

By the end of Year 3, the project will deliver:

- A production-ready multi-agent safeguarding infrastructure
- Deployment packages, documentation, and integration toolkits suitable for commercial adoption
- A defined commercialization and technology transfer plan aligned with SME product roadmaps

Figure 3 summarizes the overall 3-year development timeline, key milestones, and progression from prototype development to commercialization readiness.

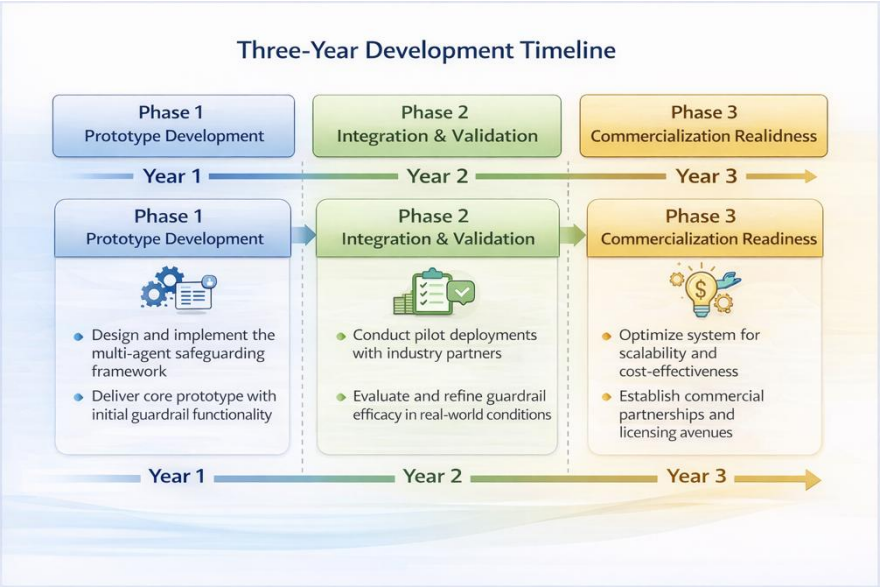

Figure 3. Three-year development timeline summarizing key milestones and the progression from core prototype development to industry validation and commercialization readiness.

Expected Outcomes

By the conclusion of the project, the joint R&D effort will deliver a deployable multi-agent AI safeguarding platform that enables safe, reliable, and scalable deployment of large AI models in industrial settings. The system will provide modular safety coverage across multiple harm categories, efficient runtime performance, and transparent coordination suitable for regulatory and enterprise environments.

This R&D plan emphasizes engineering deliverables, pilot validation, and commercialization readiness, ensuring that the developed technology transitions effectively from research concept to industry-ready safeguarding infrastructure.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the technical lead and primary developer of the proposed multi-agent AI safeguarding technology. Its main role is to provide the core scientific, engineering, and system-level expertise required to design, implement, and validate a deployable safeguarding infrastructure suitable for industrial adoption. The institute will lead the overall system architecture and technical design of the safeguarding framework, including the development of the safety orchestrator, specialized guardrail agents, and decision fusion mechanisms. It will be responsible for translating advanced research concepts in multi-agent systems and AI safety into practical, production-oriented software components that can be integrated into real-world AI inference pipelines. The research institute will conduct the applied research necessary to evaluate safeguarding accuracy, latency impact, robustness, and scalability across representative workloads. This includes designing benchmarking protocols, defining evaluation metrics, and performing systematic testing to ensure the technology meets target performance specifications and deployment requirements. The research scope will remain focused on engineering feasibility, system reliability, and commercialization readiness rather than fundamental theoretical research. The institute will develop and maintain reference implementations, prototype systems, and integration toolkits to support adoption by industry partners. It will provide technical documentation,

	deployment guidelines, and training materials to facilitate technology transfer and practical use within commercial environments. The research institute will provide the required research personnel, computational resources, and development infrastructure for the project. The principal investigator, graduate students, and dedicated research engineers will contribute to system design, implementation, testing, and documentation. The institute will also provide access to GPU-based computer platforms, software development environments, and benchmarking tools needed for large-model safeguarding experiments. Finally, the research institute will play a central role in knowledge transfer and commercialization support. It will collaborate closely with the SME partner to refine system requirements, support pilot deployments, and define commercialization pathways, including licensing strategies, product integration plans, and potential spin-off opportunities.
Expected R&R of Korean SME Partner	The Korean SME partner will serve as the primary industrial collaborator and commercialization driver for the proposed joint R&D project. Its main role is to provide real-world application contexts, development environments, and product integration pathways to ensure that the proposed safeguarding technology addresses practical industry needs and can be efficiently transitioned into commercial products. The SME partner will lead the definition of industry-relevant use cases and application scenarios where safe and reliable AI deployment is critical. This includes identifying target products, operational workflows, and risk profiles in areas such as customer service, enterprise analytics, decision-support systems, or AI-powered platforms. These use cases will guide system requirements, performance targets, and integration constraints throughout the project. The SME partner will take primary responsibility for system integration and product-oriented development tasks. This includes adapting application interfaces, configuring deployment environments, and embedding the safeguarding infrastructure into existing AI inference pipelines and production workflows. The partner will contribute engineering resources to ensure compatibility with its software stack, cloud or on-premise infrastructure, monitoring systems, and operational tools. The SME partner will play a central role in testing, validation, and performance assessment under real-world operating conditions. This includes conducting pilot deployments, collecting operational metrics, and evaluating safeguarding accuracy, latency impact, system stability, and user experience. The partner will provide structured feedback on system behavior, usability, and robustness to support iterative refinement toward production readiness. The SME partner will commit development personnel, application environments, datasets, and deployment infrastructure required for pilot testing and validation. Dedicated engineers, product managers, and quality assurance staff will participate in joint development, integration, and evaluation activities, ensuring close alignment between technical development and product timelines. Finally, the SME partner will lead commercialization planning and market deployment efforts. This includes defining product offerings incorporating the developed technology, identifying target customer segments, preparing go-to-market strategies, and supporting early customer adoption. The partner will collaborate closely with the research institute to explore licensing, product integration, and potential spin-off opportunities, ensuring that the technology transitions from prototype to revenue-generating products within the project timeframe.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** The primary technological outcome of this joint R&D project will be a deployable multi-agent AI safeguarding infrastructure that enables safe, reliable, and scalable deployment of large language models in real-world applications. The project will deliver a production-ready software system capable of detecting and mitigating key risk types, including toxicity, bias, misinformation, and privacy violations, while maintaining low latency and high system robustness. Key technical deliverables include a modular safeguarding framework with a central safety orchestrator, specialized guardrail agents, real-time monitoring and intervention mechanisms, and validated performance benchmarks demonstrating safeguarding accuracy, latency impact, and system scalability. These outcomes establish a practical foundation for trustworthy AI deployment across diverse application domains. **Commercialization Outcomes** A central outcome of the project is the transition of the developed technology from prototype to commercial deployment. Through close collaboration with the Korean SME partner, the project will deliver validated pilot systems and a production-ready safeguarding platform suitable for integration into commercial AI products and services. Expected commercialization results include incorporation of the technology into the SME partner's product portfolio, development of market-ready AI solutions with enhanced safety and reliability, and establishment of licensing or product integration pathways. These outcomes create direct opportunities for product differentiation, new revenue streams, and expansion into customer segments that require compliant and trustworthy AI systems. The project also lays the groundwork for broader commercialization channels, including technology licensing to additional SMEs, adoption by AI service providers, and potential startup or spin-off activities centered on AI safety infrastructure. **Industrial Impact** The project addresses a critical barrier to industrial AI adoption: the lack of practical, scalable, and deployable safeguarding technologies suitable for production environments. By lowering technical complexity and enabling reliable AI behavior, the proposed technology supports broader deployment of AI across sectors such as customer service, enterprise software, analytics, healthcare, and decision-support systems. Expected industrial impacts include: • Enabling safer deployment of AI systems in regulated and high-risk application domains. • Improving competitiveness of SMEs by allowing them to offer trustworthy AI-powered products. • Reducing operational risk, compliance burden, and reputational exposure associated with unsafe AI behavior. These impacts contribute to the development of a more reliable and responsible AI ecosystem accessible beyond large technology firms. **Economic and Ecosystem Impact**

	The project contributes to economic growth and innovation by strengthening the AI infrastructure capabilities of Korean SMEs and supporting the creation of high-value AI products and services. By enabling safe and reliable AI deployment, the technology lowers entry barriers for AI-driven businesses and promotes sustainable job creation in AI engineering, software development, and digital services. At the ecosystem level, the project fosters a sustainable academia–industry collaboration model that accelerates technology transfer from research to market. The outcomes strengthen national competitiveness in AI safety and infrastructure technologies and support the development of a robust, globally relevant AI innovation ecosystem. **Long-Term Impact** Beyond the project horizon, the developed safeguarding infrastructure establishes a long-term platform for trustworthy AI deployment applicable to future generations of large and multimodal models. The technology can be extended to new application domains, regulatory frameworks, and international markets, ensuring lasting relevance as AI systems continue to increase in scale and societal impact. Overall, the joint R&D project delivers both a commercially viable safeguarding technology and a strategic contribution to responsible AI adoption, enabling sustained industrial innovation and economic impact.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization